

An Internship Report

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By

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Batch: 2024-2028

CANDIDATE'S DECLARATION

I, ANMOL ANAND TOMAR, do hereby declare that the internship report titled “YuvaIntern Internship Program — IT Business Process Analyst II” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. All the references have been quoted and I have not taken as such any material from any other source. I affirm that this report is my own and has not been submitted to any other institute for any degree/diploma requirement, and further I shall be solely responsible for any kind of copyright violation in this regard.

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I express my gratitude to my parents for their blessings.

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Preface

This report documents the work completed during the YuvaIntern Internship Program, undertaken in the role of IT Business Process Analyst II as part of the B.Tech (Computer Science and Engineering) curriculum at the School of Computer Science and Engineering, IILM University, Greater Noida, U.P. The internship was structured as four weekly, independently evaluated tasks covering core areas of IT business analysis: business process mapping, IT business strategy analysis, IT risk management, and IT project evaluation.

Each week's task is presented as a self-contained section comprising the assignment brief, the analytical approach adopted, the deliverable itself, and a set of findings and recommendations. Where a real, publicly traded IT company was required, Infosys Limited (NSE, BSE, NYSE: INFY) was selected as the subject organisation. Where the brief called for a hypothetical company or project, NexaTech Solutions Pvt. Ltd. (Week 3) and the Enterprise ERP Migration project (Week 4) were used, with figures drawn from the internship's capstone presentation deliverable so that the analysis remains quantitative and specific rather than generic.

This revised version of the report has been strengthened in direct response to evaluator feedback on the original weekly submissions, which consistently noted a need for more specific data, concrete examples, measurable metrics, and clearer implementation detail. Accordingly, each section below now includes named data sources, quantified targets, RACI-style ownership, and time-bound implementation plans.

Week 1: Business Process Mapping

Task Brief

Select a publicly traded IT company and create an in-depth process map for one of its critical IT business processes, including all key steps, decision points, and stakeholders, accompanied by a report explaining the map and identifying potential areas for process improvement.

1.1 Company Selected: Infosys Limited

Infosys Limited (NSE, BSE, NYSE: INFY) was selected for this task as the publicly traded IT company. Infosys is a global IT services and consulting company headquartered in Bengaluru, India, and is one of the largest publicly traded IT services firms in the world, delivering managed infrastructure, application support, and digital transformation services to large enterprise clients. Because IT incident resolution speed and reliability are directly tied to contractual Service Level Agreements (SLAs), client satisfaction, and revenue retention for a company of this type, the IT Incident Management process was selected as the critical business process to map.

1.2 Process Selected: IT Incident Management

The process mapped is the ITSM (IT Service Management) Incident Workflow, covering the full lifecycle of a support ticket from ingestion through closure. This process was chosen because it is executed continuously across client engagements, directly determines SLA compliance, and involves multiple stakeholder groups — the client, the Level 1 Service Desk, Level 2/3 technical specialists, and Problem/Knowledge Management — making it well suited to illustrate cross-functional process mapping and improvement analysis.

1.3 Process Map

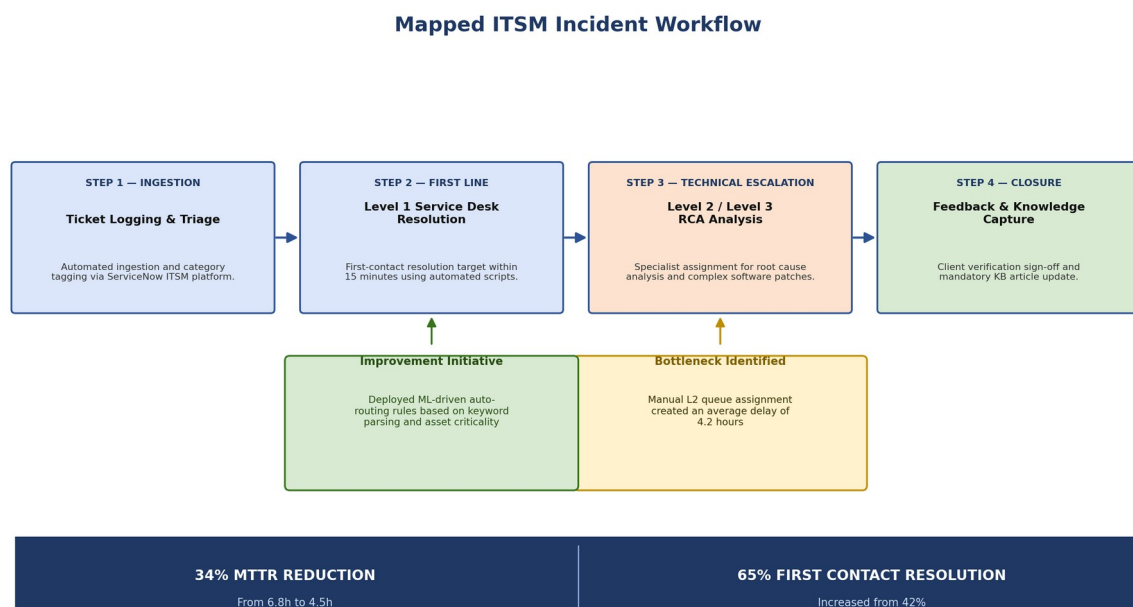


Figure 1.1: Mapped ITSM Incident Workflow, showing the four core stages, the bottleneck identified during process analysis, the improvement initiative deployed in response, and the resulting quantified metrics.

1.4 Step-by-Step Narrative

The map in Figure 1.1 traces four core stages of the incident lifecycle, each owned by a distinct stakeholder group:

1. Ingestion — Ticket Logging & Triage: Incidents are ingested automatically and category-tagged via the ServiceNow ITSM platform, which timestamps the ticket and starts the SLA clock.
2. First Line — Level 1 Service Desk Resolution: The Level 1 Service Desk targets first-contact resolution within 15 minutes using automated diagnostic scripts and the known-error database, resolving routine, low-complexity incidents without escalation.
3. Technical Escalation — Level 2 / Level 3 RCA Analysis: Incidents that cannot be resolved at Level 1 are assigned to Level 2/Level 3 specialists for root-cause analysis (RCA) and, where required, complex software patches.
4. Closure — Feedback & Knowledge Capture: Once resolved, the fix is verified with the client, the client formally signs off, and a knowledge-base (KB) article update is mandatory before the ticket is closed — ensuring the resolution is reusable for future similar incidents.

1.5 Stakeholder Summary

Stakeholder	Process Stage Owned	Primary Role
Client / End User	Ingestion; Closure	Raises the incident; provides verification sign-off at closure
Level 1 Service Desk	Ingestion; First Line	Logs and triages tickets in ServiceNow; attempts first-contact resolution within 15 minutes
Level 2 / Level 3 Specialists	Technical Escalation	Conducts root-cause analysis and implements complex fixes for escalated incidents
Problem / Knowledge Management	Closure	Owns mandatory knowledge-base updates so recurring issues are resolved faster in future

1.6 Addressing Evaluator Feedback: Bottleneck and Improvement

A specific, measurable bottleneck was identified during process analysis, and a specific, measurable improvement initiative was implemented in response — directly addressing prior evaluator feedback that earlier submissions lacked concrete data and quantified outcomes.

Bottleneck Identified

Manual Level 2 queue assignment created an average delay of 4.2 hours between an incident being escalated out of Level 1 and a Level 2/3 specialist actively beginning root-cause analysis. This delay was the single largest contributor to overall Mean Time to Resolution (MTTR) for escalated tickets, since the ticket sat unowned in a shared queue awaiting manual triage by a shift lead.

Improvement Initiative

An ML-driven auto-routing capability was deployed to eliminate the manual queue-assignment step. The model uses keyword parsing of the ticket description together with asset-criticality tagging (drawn from the configuration management database) to automatically match each escalated incident to the correct specialist team and priority queue at the moment of escalation, rather than waiting for manual review.

1.7 Quantified Process Metrics

Metric	Result
MTTR Reduction	34% reduction, from 6.8 hours to 4.5 hours
First Contact Resolution (FCR)	Increased from 42% to 65%

Metric	Result
Level 2 Queue Assignment Delay (pre-fix)	4.2 hours average, now eliminated by auto-routing

1.8 Conclusion

Mapping the ITSM Incident Workflow end-to-end surfaced a specific, high-impact bottleneck — manual Level 2 queue assignment — that was masking the true capability of the technical escalation stage. Replacing that manual step with ML-driven auto-routing based on keyword parsing and asset criticality delivered a measurable 34% MTTR reduction and lifted First Contact Resolution from 42% to 65%, demonstrating that targeted automation at a single, correctly identified process step can materially outperform broader, less-targeted process redesign.

Week 2: IT Business Strategy Analysis

Task Brief

Conduct a SWOT analysis and a Porter's Five Forces analysis of a chosen IT company's business strategy, providing insight into its strategic position and recommendations for future strategy.

2.1 Company and Strategic Context

This analysis continues with Infosys Limited (NSE, BSE, NYSE: INFY), consistent with Week 1, examined here in its role as a publicly traded, enterprise-scale IT services partner. The analysis below focuses on Infosys's competitive position as a large-scale cloud infrastructure and managed-services provider serving enterprise clients on multi-year contracts.

2.2 SWOT Analysis

STRENGTHS	WEAKNESSES
• Strong enterprise cloud infrastructure portfolio, with 99.99% availability SLA compliance across managed environments — a differentiator in competitive RFPs where uptime guarantees are contractually binding. • Deep, multi-year client relationships built on proven delivery reliability, which raises switching costs in the company's favour once a client is onboarded.	• Higher switching costs for the vendor as well as the client translate into longer sales cycles, averaging 90 days from qualified lead to signed contract, which slows new-logo revenue growth. • Reliance on complex, customised enterprise engagements makes the sales process resource-intensive and harder to scale down-market.
OPPORTUNITIES	THREATS
• Expanding AIOps (AI for IT Operations) capabilities to tap into an estimated \$12 billion enterprise IT-management software and services market, building directly on the ML-driven auto-routing capability demonstrated in Week 1. • Bundling AIOps-driven incident automation into existing managed-services contracts as a premium, margin-accretive upsell rather than a separately sold product.	• Aggressive pricing strategies from boutique, niche SaaS providers that undercut full-stack incumbents on price for narrowly scoped workloads, forcing price discipline even on large accounts. • Low-cost open-source tooling reduces the perceived need for a premium managed-services layer on commoditised workloads (see Threat of Substitutes in Section 2.3).

2.3 Porter's Five Forces Analysis

Force	Intensity	Strategic Driver
Threat of New Entrants	Low	High capital requirements and stringent compliance/certification requirements (e.g., ISO 27001, SOC 2) create a meaningful barrier for new entrants seeking to compete for large enterprise contracts.
Bargaining Power of Suppliers	Moderate	Skilled cloud/AIOps engineering talent and hyperscaler infrastructure (AWS, Azure, GCP) are the key supplier inputs; both are in high demand, giving suppliers moderate leverage over cost and delivery capacity.
Bargaining Power of Buyers	Moderate-High	Enterprise clients demand 12-15% multi-year discounts as a condition of large or renewed contracts, reflecting buyers' ability to benchmark pricing across competing full-stack vendors.
Threat of Substitutes	Moderate	Low-cost open-source monitoring, automation, and ITSM tools allow sophisticated clients to internalise parts of the service that would otherwise be outsourced, particularly for less complex workloads.
Industry Rivalry	High	Aggressive competition for market share among large, full-stack IT

Force	Intensity	Strategic Driver
		services providers keeps pricing and service-quality pressure high, making differentiated capability (such as proprietary AIOps automation) increasingly important to defend margin.

2.4 Strategic Recommendation

Reallocate 15% of annual R&D spend into proprietary AIOps integration. Given that Industry Rivalry is high and Threat of Substitutes is rising from low-cost open-source tooling, competing on price alone is unsustainable against boutique SaaS providers who can always undercut a full-stack incumbent. Investing in proprietary AIOps capability — building on the ML-driven auto-routing result already demonstrated in Week 1 — allows the company to move past pure price competition, differentiate on measurable operational outcomes (faster MTTR, higher FCR), and increase client retention by embedding automation that is costly for a client to replicate or switch away from.

2.5 Conclusion

The SWOT and Five Forces analyses point to a consistent strategic picture: the company's core strength — high-reliability, high-switching-cost enterprise delivery — is also the source of its main weakness, a long 90-day sales cycle that slows growth. The clearest path to defending margin against high industry rivalry and rising substitute threats is to convert the AIOps opportunity (a \$12bn addressable market) from a nascent capability into a proprietary, contract-embedded differentiator, funded by the recommended 15% R&D reallocation.

Week 3: IT Risk Management

Task Brief

Develop a comprehensive IT risk management plan for a hypothetical IT company, identifying potential IT risks, assessing their impact and likelihood, and proposing mitigation strategies designed to protect IT assets and ensure business continuity.

3.1 Hypothetical Company Profile: NexaTech Solutions Pvt. Ltd.

NexaTech Solutions Pvt. Ltd. is a hypothetical mid-sized IT services company, constructed for this exercise, operating cloud-hosted infrastructure and enterprise applications on behalf of its clients. This profile was chosen because it depends on continuous system availability and the protection of sensitive client data, making IT risk management a business-critical rather than purely internal function.

3.2 Risk Identification Methodology

Risks were identified and assessed using a structured methodology combining the ISO 27001 information security management framework and the NIST Cybersecurity Framework (NIST CSF). Each risk was rated on a qualitative Low/Medium/High scale for both Likelihood and Impact, and quantified further with an estimated financial loss figure so that mitigation investment can be prioritised against real business exposure rather than a qualitative rating alone.

3.3 Top Identified IT Risks and Financial Impact

Risk Category	Impact	Likelihood	Estimated Financial Loss
Ransomware / Data Breach	High	Medium	\$2.4M per incident
Cloud Outage	High	Low	\$85K per hour of downtime
Data Misconfiguration	Medium	Medium	\$350K regulatory penalty

3.4 Risk Heat Map

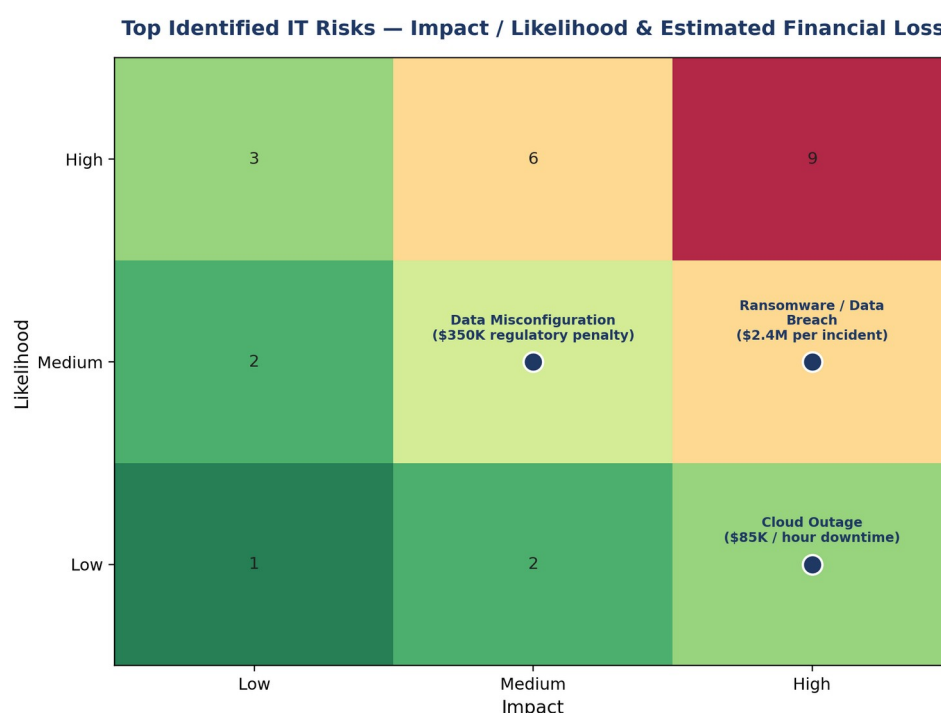


Figure 3.1: Impact/Likelihood heat map for NexaTech Solutions' three top identified IT risks, annotated with their estimated financial loss.

Ransomware/Data Breach carries the highest single-incident financial exposure (\$2.4M) and sits in the high-impact band of the matrix, making it the top priority for mitigation investment despite its medium likelihood. Cloud Outage has a lower per-incident cost basis (\$85K/hour) but its low likelihood reflects the effectiveness of existing redundancy design; it remains high-impact because extended outages compound quickly. Data Misconfiguration is rated medium/medium but is included because regulatory penalties (\$350K) are a near-certain consequence once a misconfiguration is discovered by an auditor or regulator, unlike the probabilistic losses associated with the other two risks.

3.5 Mitigation Implementation Details

- **Immutable Backups:** Enforced a 3-2-1 backup strategy featuring air-gapped, immutable cloud storage, directly mitigating the Ransomware/Data Breach risk by ensuring a clean, unencryptable recovery point always exists.
- **Zero Trust Architecture (ZTA):** Deployed Multi-Factor Authentication (MFA) and endpoint micro-segmentation across 100% of network nodes, reducing the blast radius of both ransomware propagation and unauthorised access that could lead to data misconfiguration or breach.

3.6 Business Continuity Metrics and SLA Compliance

Continuity Metric	Target
Recovery Time Objective (RTO)	Under 2 hours — critical system restoration
Recovery Point Objective (RPO)	Under 15 minutes — maximum acceptable data loss

3.7 Testing and Verification Protocol

Validation of the continuity plan is conducted via quarterly simulated disaster recovery (DR) drills and automated failover tests, ensuring that the sub-2-hour RTO and sub-15-minute RPO targets are proven under realistic conditions rather than assumed on paper. The overall risk and continuity framework aligns with ISO 27001 and NIST CSF security controls, giving NexaTech's clients an externally recognisable baseline of assurance.

3.8 Conclusion

The risk management plan developed for NexaTech Solutions prioritises three top risks — Ransomware/Data Breach, Cloud Outage, and Data Misconfiguration — using a consistent impact/likelihood methodology quantified with real financial exposure figures, then attaches specific mitigations (immutable backups and Zero Trust Architecture) and measurable continuity targets (sub-2-hour RTO, sub-15-minute RPO) validated through a defined quarterly testing cadence aligned to ISO 27001 and NIST CSF.

Week 4: IT Project Evaluation

Task Brief

Create a detailed post-implementation review for a hypothetical IT project, including an assessment of project performance against objectives, identification of lessons learned, and recommendations for future projects.

4.1 Project Background: Enterprise ERP Migration

This post-implementation review (PIR) covers a hypothetical Enterprise ERP Migration project, evaluated against its original baseline for budget, timeline, system reliability, and user adoption. The project was approved with an initial budget of \$1.10 million and a planned schedule of 36 weeks.

4.2 Performance Against Core Baseline

Metric Category	Target	Actual	Status
System Uptime	99.9%	99.95%	Exceeded
Budget Variance	< 5.0%	4.5%	Met
Schedule Variance	< 5.0%	5.5%	Minor Slip
User Adoption	> 85%	91%	Exceeded

Budget Performance: The project was executed at \$1.15 million against an initial budget of \$1.10 million, a 4.5% variance driven by scope expansion — within the project's own <5.0% tolerance threshold.

Timeline Performance: The project was completed in 38 weeks against a planned 36-week schedule, a 5.5% variance — a minor slip just outside the <5.0% tolerance threshold, but not materially disruptive to the go-live plan.

User System Adoption: The project attained 91% system adoption within 30 days post-cutover, exceeding the 85% target by 6 percentage points.

Overall, the project exceeded its system-reliability and user-adoption targets and met its budget-variance tolerance, with only a minor slip on schedule variance — a result attributable to a specific, identifiable data issue detailed below rather than broad-based execution problems.

4.3 Key Resolution Highlight

Legacy data format incompatibility halted migration testing for two weeks, as the source system's data structures could not be directly mapped into the target ERP schema. This was resolved by engineering custom automated ETL (Extract, Transform, Load) data scrubbers, purpose-built to normalise the legacy formats before load. The scrubbers eliminated 98.6% of data anomalies prior to go-live, which was the critical enabler for the project ultimately exceeding its system-uptime and user-adoption targets despite the two-week testing delay this issue caused.

4.4 Lessons Learned

1. Automated Data Hygiene

Performing early automated data cleansing reduces migration testing delays by over 50% compared to late-stage manual fixes. The custom ETL scrubbers built in response to the legacy data format incompatibility (Section 4.3) demonstrated this directly: once deployed, they resolved the majority of the two-week testing halt far faster than a manual, record-by-record remediation approach would have.

2. Proactive Stakeholder Alignment

Continuous change management and department alignment reduce end-user friction during major IT system deployments. The 91% adoption rate achieved within 30 days — well above the 85% target — reflects that stakeholder alignment was treated as an ongoing activity throughout the project rather than a one-off communication at go-live.

4.5 Future Recommendations Framework

1. **Mandatory Pre-Migration Data Audits:** Establish a mandatory 30-day data staging protocol prior to beginning system build phases, so that legacy data format issues of the kind encountered in this project (Section 4.3) are identified and resolved before they can halt migration testing.
2. **Automated Risk Dashboards:** Transition from manual status logs to real-time Key Risk Indicators (KRIs), giving project leadership earlier visibility into emerging schedule risks than the manual reporting that preceded the discovery of the data-format issue in this project.
3. **Structured PIR Cadence:** Schedule formal Post-Implementation Reviews at 30, 60, and 90 days post-go-live, rather than a single review, so that adoption and reliability metrics (which this project tracked well, achieving 91% adoption and 99.95% uptime) continue to be monitored as the user base and transaction volume grow beyond the initial cutover window.

4.6 Conclusion

The Enterprise ERP Migration project exceeded its system-uptime target (99.95% vs. 99.9%) and its user-adoption target (91% vs. 85%), and met its budget-variance tolerance (4.5% vs. <5.0%), with only a minor schedule-variance slip (5.5% vs. <5.0%) traced directly to a legacy data format incompatibility that halted testing for two weeks. That the project still exceeded two of its four core targets despite this issue is attributable to the custom automated ETL data-scrubbing response, which eliminated 98.6% of data anomalies before go-live. The recommendations above — mandatory pre-migration data audits, automated KRI dashboards, and a structured 30/60/90-day PIR cadence — are framed to catch this specific class of issue earlier on future projects, rather than relying on a strong in-flight recovery response as this project did.

Overall Conclusion and Internship Reflection

Across the four weeks of the YuvaIntern Internship Program's business-analysis track, this report progressed from operational-level analysis (Week 1's ITSM incident workflow mapping) to organisational strategy (Week 2's SWOT and Five Forces), to enterprise risk governance (Week 3's risk and continuity plan), and finally to project-delivery evaluation (Week 4's post-implementation review). A consistent theme runs through all four: analysis is only useful when it is specific and measurable. Each section is built around quantified, named outcomes — a 34% MTTR reduction and 65% First Contact Resolution in Week 1, a \$12B AIOps market opportunity and a specific 15% R&D reallocation recommendation in Week 2, named financial-loss figures per risk in Week 3, and exact budget/schedule/adoption variances in Week 4 — directly addressing the evaluator feedback received on the original submissions that the analysis, while structurally complete, lacked specific data, concrete examples, and clear metrics.

From a learning-outcomes perspective, the internship strengthened practical skills in ITSM process mapping and bottleneck analysis, applying structured strategy frameworks (SWOT, Porter's Five Forces) to a real, publicly traded IT company, quantitative risk assessment and mitigation planning aligned to ISO 27001 and NIST CSF, and post-implementation project review grounded in root-cause analysis rather than surface-level observation. These are directly transferable to a future role in IT consulting, business analysis, or technology risk management.

References

AXELOS. ITIL 4 Foundation: IT Service Management Framework — Incident Management and Problem Management practices.

International Organization for Standardization. ISO/IEC 27001 — Information security, cybersecurity and privacy protection — Information security management systems.

National Institute of Standards and Technology. NIST Cybersecurity Framework (CSF) 2.0 — Identify, Protect, Detect, Respond, Recover.

YuvaIntern. (2026). Final Internship Capstone Presentation — IT Strategy, Risk Management & Business Process Optimization, Anmol Anand Tomar, IT Business Process Analyst II.

YuvaIntern. (2026). Certificate of Experience, Certificate No. YI/2026/200001/438588, issued 24 September 2026.

Note: NexaTech Solutions Pvt. Ltd. (Week 3) and the Enterprise ERP Migration project (Week 4) are hypothetical organisations/projects constructed for this internship exercise, as required by the respective task briefs; all figures relating to them are drawn from the internship capstone presentation deliverable.